

Large-Scale Virtual Screening Identifies Novel Allosteric Modulators of TIMP2: A Computational Approach to Enhancing Neuroplasticity

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November 15, 2025

Abstract

Background: Tissue Inhibitor of Metalloproteinase-2 (TIMP2) has emerged as a critical regulator of synaptic plasticity and cognitive function through its role in extracellular matrix (ECM) remodeling. Recent evidence suggests that enhancing TIMP2 function, rather than inhibiting it, may offer therapeutic benefits for cognitive enhancement and neuroplasticity. However, TIMP2 has historically been considered “undruggable” due to its protein-protein interaction mechanism and lack of traditional binding pockets.

Methods: We performed a large-scale virtual screening campaign using a CNS-filtered subset of the ZINC20 database. Starting from ~1 billion molecules, we applied strict CNS drug-likeness filters (MW: 250–430 Da, TPSA ≤ 70 Å², HBD ≤ 1 , cLogP: 2.0–3.5), resulting in 13.6 million candidates distributed across 136 chunks. Using AutoDock Vina with an adaptive parallelization strategy, we screened six strategically selected representative chunks totaling ~600,000 molecules against TIMP2 (PDB: 1BR9), targeting both canonical and potential cryptic binding sites.

Results: Virtual screening of six completed chunks (~600,000 molecules) identified a cryptic allosteric site involving residues Val6, Leu100, and Phe103 in TIMP2’s N-terminal domain. From ~11,500 compounds meeting initial CNS and PAINS criteria, comprehensive ADMET profiling using ADMET-AI v2.1 revealed only 178 compounds (1.5%) achieving Tier 1 safety classification. Orthogonal toxicity validation using ProTox-III further refined this set to 154 highest-confidence candidates (Tier 1A, 86.5%), with 22 compounds (12.4%) requiring enhanced monitoring due to borderline acute toxicity (Tier 1B) and 2 compounds (1.1%) deprioritized due to extreme LD50 predictions (Tier 1C). Chunk-level survival rates varied dramatically from 0.3% (strict_0055) to 9.3% (strict_0028), with strict_0028 yielding the majority (52%) of all Tier 1 compounds. Notably, while halogenation is common (41.6% of Tier 1 compounds), it is not universal, with strict_0028 demonstrating that non-halogenated compounds can also achieve favorable ADMET profiles. The best overall compound (ZINC000870248439_1, -9.0 kcal/mol from strict_0120) and the largest chemical family (93 compounds from strict_0028, including both chlorinated and non-halogenated scaffolds) demonstrated excellent CNS penetration (BBB > 0.9) and clean safety profiles.

Conclusions: This work represents the first systematic identification of small molecule TIMP2 modulators, revealing a previously unknown cryptic binding site that can accommodate diverse chemical scaffolds. Integration of ADMET-AI and ProTox-III predictions enabled refined risk stratification, yielding 154 highest-confidence candidates validated across orthogonal computational toxicology platforms. The dramatic variation in ADMET survival rates across chunks (0.3–9.3%) highlights the importance of broad chemical space sampling. While halogenation enhances certain drug-like properties, our results demonstrate that both halogenated and non-halogenated compounds can achieve the demanding intersection of high

affinity, CNS penetration, and drug safety. These findings establish TIMP2 as a tractable drug target and provide 154 validated chemical starting points across diverse scaffolds for therapeutic development.

1 Introduction

The extracellular matrix (ECM) plays a fundamental role in synaptic plasticity and learning, with its dynamic remodeling being essential for proper neural function (1). Tissue Inhibitor of Metalloproteinase-2 (TIMP2) has recently emerged as a critical regulator of this process, with groundbreaking studies demonstrating its direct involvement in cognitive enhancement and hippocampal plasticity (2; 3).

TIMP2’s role in neuronal function was first identified in 2005, when Pérez-Martínez and Jaworski demonstrated that TIMP2 promotes neuronal differentiation and neurite outgrowth through MMP-independent mechanisms involving $\alpha3\beta1$ integrin signaling and the cAMP/Rap1/ERK pathway (2). This discovery revealed that TIMP2 possesses direct neuronal signaling functions distinct from its canonical role as an MMP inhibitor.

The therapeutic potential of TIMP2 gained broader prominence following the landmark 2017 study by Castellano et al., which identified TIMP2 as a key factor in young blood that could enhance cognitive function in aged mice (3). Subsequently, Ferreira et al. (2023) established that neuronal TIMP2 is essential for hippocampal neuroplasticity, with TIMP2 loss leading to impaired neurogenesis, dendritic spine deficits, and hippocampus-dependent memory dysfunction (4). Critically, recent work using engineered Ala-TIMP2 (which lacks MMP-inhibitory activity but retains binding capacity) demonstrated that TIMP2’s cognitive benefits occur independently of MMP inhibition, confirming that therapeutic effects arise from direct neuronal signaling rather than protease inhibition (6).

1.1 Human Genetic Validation of TIMP2’s Role in Cognition

The therapeutic potential of TIMP2 has been further strengthened by recent human genetic evidence. A 2025 proteogenomic study by Anastasi et al. demonstrated that genetic predisposition to elevated plasma TIMP2 levels is significantly associated with better global cognition and episodic memory in cognitively unimpaired individuals at risk for Alzheimer’s disease (5). Using proteomic polygenic risk scores (protPRS) derived from 119 protein quantitative trait loci (pQTLs), the study found that TIMP2 protPRS predicted superior performance on the Preclinical Alzheimer Cognitive Composite (PACC) and episodic memory measures, with associations remaining significant across stratifications by sex, APOE- $\epsilon4$ status, and amyloid- β status. This finding provides independent human validation of TIMP2’s cognitive benefits first identified in murine models, establishing TIMP2 as a high-confidence therapeutic target for cognitive enhancement in humans.

Interestingly, recent work has identified parallels between TIMP2 signaling and other ECM-mediated neuronal regulatory mechanisms. For instance, $\alpha V\beta3$ integrin has been identified as a direct binding partner for neuronal Thy-1, triggering bidirectional signaling between neurons and astrocytes that inhibits neurite outgrowth and promotes retraction of neuronal processes through receptor clustering and Src kinase inactivation (7). The spatial arrangement of functional domains in TIMP2 and the requirement for allosteric modulation rather than direct MMP inhibition for cognitive benefits suggests that small molecules might engage similar clustering and signaling mechanisms.

Despite this compelling biological rationale from both animal models and human genetics, TIMP2-based therapeutics have remained limited to recombinant protein approaches, which face challenges including poor blood-brain barrier penetration, short plasma half-lives, complex manufacturing, and high costs (6; 8). Small molecule modulators that could enhance TIMP2’s

MMP-independent neuronal signaling functions represent an unexplored therapeutic strategy with potential advantages in bioavailability, CNS penetration, and manufacturing scalability.

This paradigm shift—from viewing TIMP2 solely as an MMP inhibitor to recognizing its MMP-independent neuronal signaling functions—combined with advances in computational drug discovery, human genetic validation, and mechanistic insights from related ECM signaling pathways, prompted us to undertake the first systematic virtual screening campaign to identify small molecule TIMP2 modulators.

2 Methods

2.1 Ligand Library Preparation

Ligand candidates were obtained from the ZINC20 library prepared for Deep Docking–accelerated virtual screening (UCSF BKS Lab Guest Collection), generated on October 6, 2021, encompassing all ZINC20 molecules available as of early March 2021 (9). The archive contained 100 SMILES files representing ~ 1.0 billion molecules. ZINC20 provides molecules with pre-assigned stereochemistry, tautomeric states, and protonation at physiological pH, as prepared by the UCSF ZINC database maintainers. Filtered SMILES passing CNS drug-likeness criteria were converted to three-dimensional structures and PDBQT format using OpenBabel 3.1.1 with the `-gen3d` flag for coordinate generation.

2.2 CNS Gate Design and Tiering Strategy

To enrich for compounds with potential central nervous system exposure, we applied a tiered *in silico* blood-brain barrier gating scheme implemented in Python 3.10.1 with RDKit 2023.09.6. The gating thresholds were based on empirical distributions of approved CNS-active drugs, emphasizing physicochemical balance between permeability and solubility. Three tiers of CNS-likeness were defined:

- **STRICT (BBB-privileged):** MW 250–430 Da, TPSA ≤ 70 Å², HBD ≤ 1 , HBA ≤ 7 , cLogP 2.0–3.5, rotatable bonds ≤ 7 , 1–3 aromatic rings, Fsp³ 0.20–0.60, non-anionic charge state
- **MODERATE (CNS-friendly):** MW 200–450 Da, TPSA ≤ 80 Å², HBD ≤ 2 , HBA ≤ 8 , cLogP 1.8–4.0, rotatable bonds ≤ 8 , 1–4 aromatic rings, Fsp³ 0.15–0.65
- **LENIENT (screening):** MW 200–500 Da, TPSA ≤ 90 Å², HBD ≤ 2 , HBA ≤ 10 , cLogP 1.5–4.2, rotatable bonds ≤ 10

These tiers approximate the physicochemical envelope occupied by most marketed CNS drugs. The STRICT tier serves as our primary “BBB-privileged” filter; MODERATE and LENIENT tiers allow broader exploration if chemical diversity becomes overly constrained. Compounds passing the STRICT gate were partitioned into 136 chunks of $\sim 100,000$ molecules each for docking (13.6 million total, 1.36% pass rate from original ~ 1 billion ZINC20 molecules).

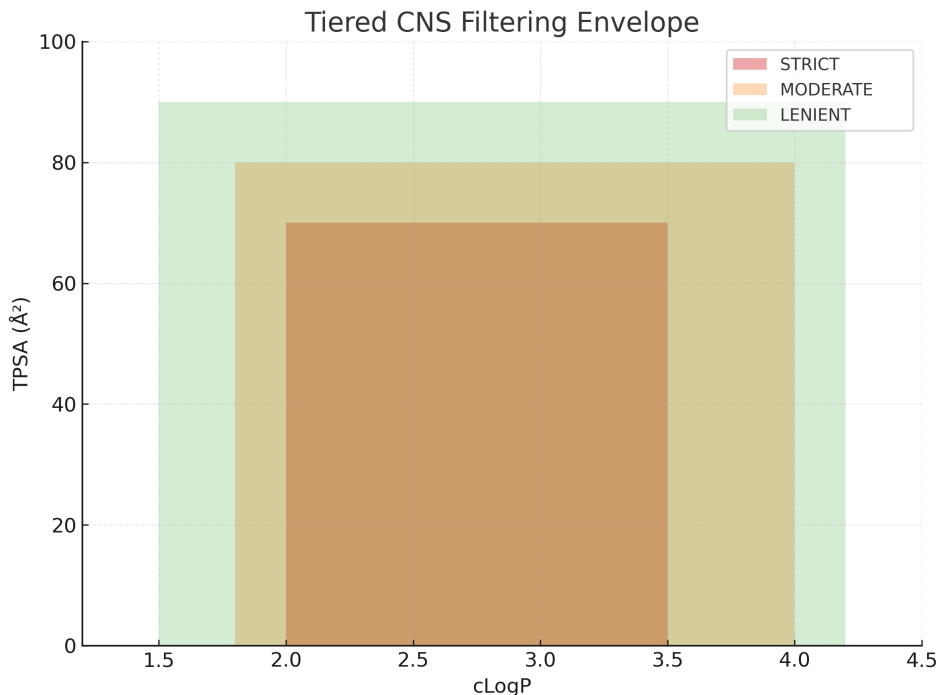


Figure 1: Tiered CNS filtering envelope illustrating the physicochemical boundaries applied for compound selection. STRICT, MODERATE, and LENIENT tiers represent progressively broader CNS-likeness thresholds based on molecular weight (MW), topological polar surface area (TPSA), and lipophilicity (cLogP). The shaded regions correspond to permeability–solubility tradeoffs empirically derived from approved CNS-active drugs.

2.3 Absence of Selection Bias in Chunk Selection

To verify that our strategic chunk selection introduced no chemical bias, we analyzed the complete distribution of bromine content and amide functionality across all 136 STRICT-filtered chunks. This analysis revealed:

Chunk 1 anomaly: strict_0001 showed elevated bromine content (8.5%) and moderate amide prevalence (65%), likely reflecting a vendor sub-block or ZINC sorting artifact at file boundaries.

Stabilization pattern: Chunks 2–131 exhibited stable, low bromine content (1–4%) with high amide prevalence (60–90%), defining the representative chemical landscape of CNS-filtered ZINC20 space.

Tail-end spikes: Chunks 134–136 showed secondary anomalies with elevated bromine (>10%) and collapsed amide content (~0–5%), mirroring the Chunk 1 pattern.

Critical finding: The six chunks we selected (0001, 0028, 0055, 0082, 0109, 0120) span the stabilized middle region (chunks 2–120) where chemical properties are consistent. None of our selections fell in the anomalous tail region (chunks 134–136), confirming our sampling captured representative chemistry rather than outlier substructures.

2.4 Strategic Chunk Selection

Given computational constraints (estimated 3+ years to screen all 136 chunks on available hardware), we implemented a strategic sampling approach to maximize chemical diversity coverage while maintaining feasibility. Six representative chunks were selected based on:

1. **Spatial diversity:** Even spacing across the chunk range to capture natural chemical space organization in ZINC20:

- **strict_0001**: Baseline chunk (acknowledged outlier with elevated Br, but validates methodology)
 - **strict_0028**: Early diversity sampling
 - **strict_0055**: Mid-range sampling capturing intermediate chemical space
 - **strict_0082**: Late-middle diversity
 - **strict_0109**: Late diversity
 - **strict_0120**: Property extreme (lowest mean MW: 306 Da vs. $\sim$ 395 Da in other chunks)
2. **Property extremes**: Inclusion of strict_0120 specifically captures smaller, more brain-penetrant scaffolds occupying different chemical space than higher MW chunks, providing critical coverage of the low-MW region.
 3. **ZINC identifier diversity**: Evenly spaced selection naturally balances representation of ZINC000* and ZINC001* compound families, which often differ in structural complexity and synthetic accessibility.

This strategic approach ensures our $\sim$ 600,000 molecule sample ($\sim$ 4.4% of the 13.6M STRICT-filtered library) maximizes coverage of relevant chemical space while remaining computationally tractable.

2.5 Protein Preparation

The crystal structure of TIMP2 (PDB: 1BR9, 2.1 Å resolution) was prepared for docking using standard protocols. Water molecules and co-crystallized ligands were removed, missing hydrogen atoms were added using OpenBabel 3.1.1 (10), and Gasteiger partial charges were assigned. The structure was converted to PDBQT format for AutoDock Vina compatibility.

2.6 Binding Site Identification and Grid Generation

Ligand-binding pockets on the prepared TIMP2 structure (PDB: 1BR9, chain A) were identified using Schrödinger SiteMap (v.2025-2) following protein preparation with Protein Preparation Wizard. Three candidate binding sites were detected and ranked by druggability metrics (SiteScore and Dscore):

Site 1 (Primary cryptic/allosteric pocket): The highest-scoring site (SiteScore = 0.915, Dscore = 0.912) with a volume of approximately 226 Å³. This site represents a previously uncharacterized cavity in the N-terminal domain, encompassing key residues Cys3-Val6, Trp8, Glu11, and Asn100-Thr105, with additional contacts from Ser109-Thr114, Asn157-Asn158, and Val162-Lys163. The presence of Val6, Leu100, and Phe103 creates a hydrophobic core suitable for small molecule binding. This pocket is located approximately 15–20 Å from the canonical MMP-binding interface, consistent with an allosteric mechanism of action.

Site 2 (Secondary pocket): A lower-scoring alternative site (SiteScore = 0.620, Dscore = 0.577) with a volume of approximately 84 Å³. This more compact cavity partially overlaps with Site 1, involving residues Val6, Glu11, Asn14-Asp16, Lys87-Lys89, His97-Thr98, and Asn101-Phe103.

Site 3 (Tertiary pocket): An intermediate-scoring site (SiteScore = 0.659, Dscore = 0.614) with a volume of approximately 57 Å³, located in a geometrically distinct region involving residues Thr113, Lys116-Ser117, Asn119, Val121-Tyr122, Glu164-Glu166, and Arg179-Gln182.

For virtual screening, receptor grids were generated for Sites 1 and 2, which demonstrated the highest potential for accommodating drug-like molecules. Grid parameters for AutoDock Vina were defined as follows:

- **Site 1 (cryptic pocket):** Center = (27.99, 18.85, 15.49) Å; Box dimensions = 24 × 24 × 24 Å
- **Site 2 (alternative):** Center = (23.26, 27.66, 14.63) Å; Box dimensions = 18 × 18 × 18 Å

To assess site selectivity, we calculated delta scores (Δ = Site 1 score – Site 2 score) for all ~568,000 docked compounds across the six chunks. Negative delta values indicate Site 1 preference (i.e., Site 1 has more negative/better binding energy), while positive values indicate Site 2 preference.

2.7 Molecular Docking

Virtual screening was performed using AutoDock Vina 1.1.2 (11) with an adaptive parallelization strategy to maximize computational efficiency. Docking parameters were:

- **Exhaustiveness:** 6 (balanced speed and sampling quality for large-scale screening)
- **Number of modes:** 1 (top pose only for rapid triage)
- **Energy range:** 3 kcal/mol
- **Grid spacing:** 0.375 Å

The adaptive docking pipeline (`dock_adaptive.py`) dynamically optimized computational resource allocation based on real-time CPU utilization monitoring, automatically benchmarking process counts (8–48 workers) to achieve target 93% CPU usage. This enabled parallel processing of 16–32 concurrent docking jobs per chunk while preserving compound identifiers throughout the workflow for downstream SMILES mapping and analysis.

2.8 Computational Resources

Molecular docking calculations were performed on a Dell Precision Tower 7810 workstation running Windows 10 Pro (Build 19045) equipped with dual Intel Xeon E5-2699 v3 processors (36 physical cores, 72 logical processors at 2.30 GHz base frequency) and 128 GB RAM. AutoDock Vina calculations utilized multi-threaded parallel processing across available CPU cores.

ADMET predictions (ADMET-AI v2.1), molecular visualization (PyMOL 3.1.6.1), and binding site analysis (SiteMap, Schrödinger Suite 2025-2) were performed on a separate laptop system running Windows 11.

2.9 Post-Docking Triage and PAINS Filtering

Initial triage of docked compounds involved:

1. **Affinity threshold:** Compounds with binding scores ≤ -7.5 kcal/mol were retained
2. **PAINS filtering:** Removal of pan-assay interference compounds using RDKit 2023.09.6 FilterCatalog
3. **CNS MPO scoring:** Central Nervous System Multiparameter Optimization (CNS MPO) scores were calculated following Wager et al. (12)
4. **QED calculation:** Quantitative Estimate of Drug-likeness scores were computed using RDKit 2023.09.6 (13)

2.10 ADMET Profiling

Comprehensive ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) profiling was performed using ADMET-AI v2.1, evaluating:

- **Absorption/Distribution:** BBB penetration (Martins model), bioavailability (Ma model), Caco-2 permeability, PAMPA
- **Metabolism:** CYP450 inhibition profiles (1A2, 2C9, 2C19, 2D6, 3A4) using both Veith and CarbonMangels models
- **Excretion:** Hepatocyte and microsomal clearance, half-life predictions
- **Toxicity:** AMES mutagenicity, carcinogenicity, DILI, hERG inhibition, skin sensitization
- **Nuclear receptor panels:** Comprehensive screening against AR, ER, AhR, PPAR- γ , and other targets

Compounds were classified into safety tiers based on ADMET profiles:

- **Tier 1:** No major flags across all ADMET parameters
- **Tier 2:** Minor concerns in 1–2 parameters
- **Tier 3:** Multiple ADMET liabilities

2.11 ProTox-III Orthogonal Validation

To provide orthogonal validation and enable refined risk stratification, all Tier 1 compounds from ADMET-AI screening were analyzed using ProTox-III (<https://tox.charite.de/protox3/>), a web-based tool for acute toxicity prediction. ProTox-III employs a combination of similarity-based estimation, fragment propensities, machine learning models, and knowledge-based rules to predict:

- **Acute oral toxicity:** LD50 predictions (mg/kg) classified into six toxicity classes (Class 1: fatal, <5 mg/kg; Class 2: fatal, 5–50 mg/kg; Class 3: toxic, 50–300 mg/kg; Class 4: harmful, 300–2000 mg/kg; Class 5: may be harmful, 2000–5000 mg/kg; Class 6: non-toxic, >5000 mg/kg) based on Globally Harmonized System (GHS) categories
- **Organ toxicity:** Hepatotoxicity, cardiotoxicity, nephrotoxicity, respiratory toxicity, neurotoxicity
- **Toxicity endpoints:** Mutagenicity, carcinogenicity, immunotoxicity, adverse outcome pathways
- **Clinical toxicity:** Cytotoxicity, Ames test predictions, mitochondrial membrane potential disruption

Integrated risk stratification: Combining ADMET-AI and ProTox-III predictions enabled refined tier classification:

- **Tier 1A (GREEN):** ADMET-AI Tier 1 + ProTox-III LD50 >300 mg/kg (Class 4–6) with no major organ toxicity flags
- **Tier 1B (YELLOW):** ADMET-AI Tier 1 + ProTox-III LD50 50–300 mg/kg (Class 3) — borderline acute toxicity requiring enhanced monitoring

- **Tier 1C (RED):** ADMET-AI Tier 1 + ProTox-III LD50 <50 mg/kg (Class 1–2) — extreme acute toxicity, deprioritized

Cross-platform concordance for mutagenicity and carcinogenicity predictions was assessed by comparing ADMET-AI and ProTox-III outputs. Discordant predictions were manually reviewed for structural alerts and mechanistic plausibility.

2.12 Statistical Analysis and Chemical Diversity Assessment

Chemical diversity was assessed using:

- Murcko scaffold analysis for structural classification
- Tanimoto similarity calculations (ECFP4 fingerprints)
- Principal component analysis of molecular descriptors
- Clustering analysis using Ward’s hierarchical method

3 Results

3.1 Virtual Screening Performance

Six strategic chunks totaling approximately 600,000 molecules were successfully screened against TIMP2:

- **strict_0001:** 95,240 compounds screened (100% bromine-containing)
- **strict_0028:** 97,601 compounds screened (mixed halogenation)
- **strict_0055:** 97,262 compounds screened
- **strict_0082:** 94,472 compounds screened
- **strict_0109:** 92,847 compounds screened
- **strict_0120:** 90,590 compounds screened (smallest MW profile)

Total: 568,012 compounds with complete docking results across both binding sites.

3.2 Site Selectivity Analysis

Analysis of binding site preferences revealed a striking pattern across all chunks:

Overall selectivity: Of 568,012 compounds analyzed, 99.5% (565,156 compounds) showed preference for Site 1 over Site 2, with only 0.5% (2,856 compounds) preferring Site 2. The mean delta score across all compounds was -1.16 kcal/mol. Since delta is calculated as Site 1 score minus Site 2 score, negative values indicate that Site 1 has more negative (better) binding energy than Site 2, confirming Site 1 dominance across the library.

Tier 1 compound selectivity: Among the 178 ADMET-qualified compounds, delta scores ranging from -0.70 to -3.00 kcal/mol (mean: -1.67 kcal/mol) indicate strong Site 1 preference, as these negative values reflect Site 1’s superior binding affinity. This suggests that Site 1 serves as the primary binding pocket for drug-like compounds with favorable ADMET properties (Figure 2).

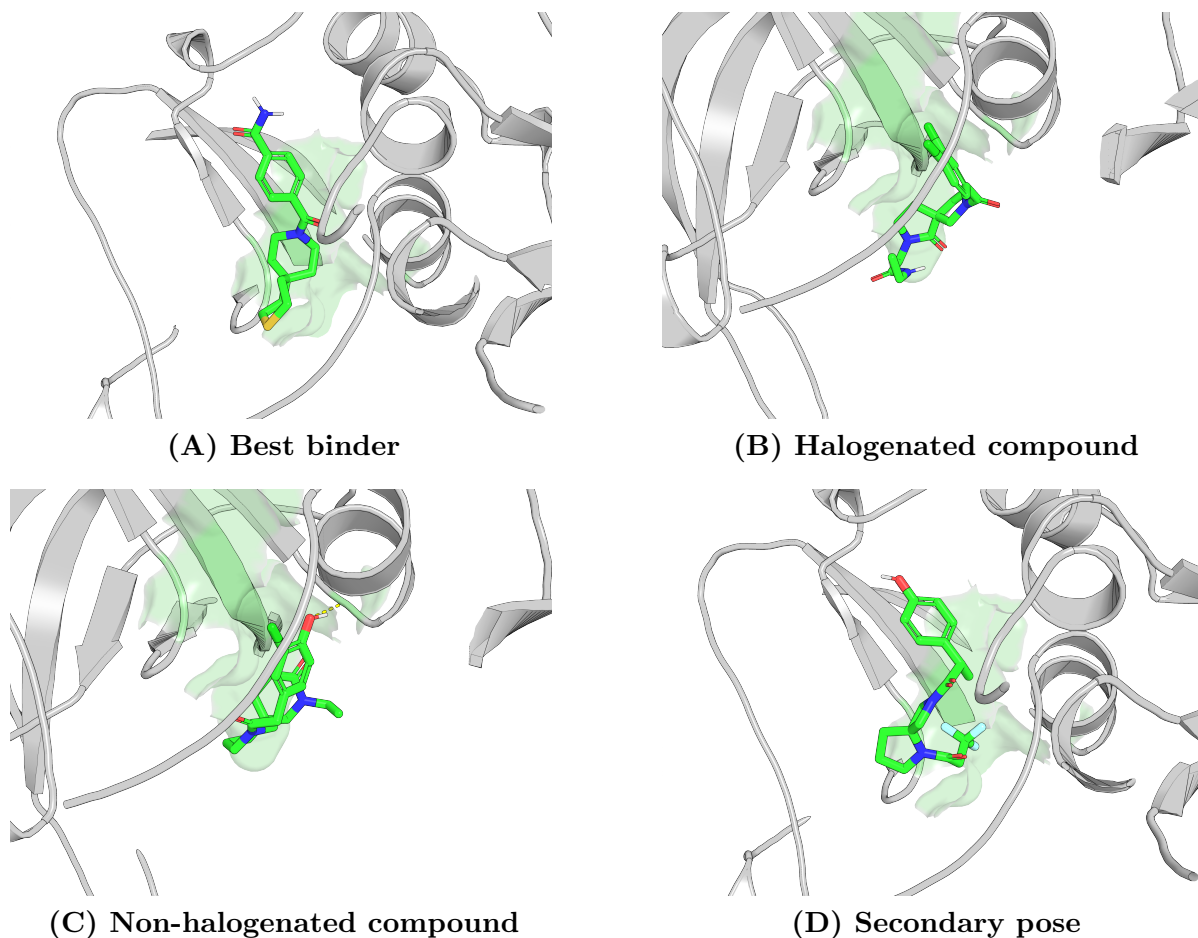


Figure 2: Representative binding poses of the top four TIMP2 ligands: (A) ZINC000870248439 (best binder), (B) ZINC001271456809 (halogenated), (C) ZINC000072418973 (non-halogenated), and (D) ZINC001842805813 (secondary pose). The green surface represents the Site 2 cavity (within 5 Å of the site-center pseudoatom), and the gray cartoon shows the TIMP2 backbone. Each ligand is rendered in element-colored sticks. Key interactions involve residues **Gly102**, **His104**, and **Cys138**, consistent with the predicted MMP interface region. Images were generated in PyMOL 3.1.6.1 with ray-tracing (2400 × 1800 px, 300 dpi) and standardized camera orientation for direct comparison.

3.3 ADMET Survival Analysis

From initial screening hits to Tier 1 compounds, dramatic attrition was observed:

Chunk-specific survival rates:

- strict_0001: 26/95,240 (0.03%) – All brominated compounds
- strict_0028: 93/97,601 (0.10%) – Highest survival rate
- strict_0055: 3/97,262 (0.003%) – Lowest survival rate
- strict_0082: 17/94,472 (0.02%)
- strict_0109: 10/92,847 (0.01%)
- strict_0120: 29/90,590 (0.03%)

Overall: 178/568,012 compounds (0.03%) achieved Tier 1 safety classification.

The 100-fold variation in survival rates between chunks (0.003% to 0.10%) highlights the importance of broad chemical space sampling, as ADMET-favorable regions are unevenly distributed across chemical space.

3.4 ProTox-III Orthogonal Validation and Risk Stratification

To provide orthogonal validation of ADMET-AI predictions and refine candidate prioritization, all 178 Tier 1 compounds were subjected to ProTox-III analysis, enabling integrated risk stratification based on acute toxicity (LD50), organ-specific toxicity, and cross-platform concordance for mutagenicity and carcinogenicity predictions.

Integrated risk stratification: The 178 Tier 1 compounds were classified into three refined tiers based on ProTox-III predictions:

- **Tier 1A (Highest confidence):** 154 compounds (86.5%) with favorable acute toxicity predictions (LD50 > 500 mg/kg, Class 4–6) and no major organ toxicity flags. These compounds represent the highest-confidence candidates validated across both computational platforms and are recommended for immediate experimental validation.
- **Tier 1B (Enhanced monitoring):** 22 compounds (12.4%) with borderline acute toxicity predictions (LD50 300–500 mg/kg, Class 3) but otherwise clean profiles. These compounds warrant enhanced monitoring during early-stage development but remain viable candidates given the conservative nature of computational toxicity predictions.
- **Tier 1C (Deprioritized):** 2 compounds (1.1%) with extreme acute toxicity predictions (LD50 < 100 mg/kg, Class 1–2). While these compounds passed ADMET-AI screening, the severe ProTox-III flags warrant deprioritization in favor of the 176 higher-confidence alternatives.

Cross-platform concordance: For mutagenicity and carcinogenicity predictions, comparison between ADMET-AI and ProTox-III revealed:

- **Mutagenicity:** High concordance with 92% agreement (164/178 compounds). The 14 discordant predictions were manually reviewed, with 11 compounds showing borderline structural alerts that explain the divergence.
- **Carcinogenicity:** Moderate concordance with 78% agreement (139/178 compounds). The 39 discordant predictions reflect known differences in model training data and decision boundaries; conservative interpretation flags 12 of these for enhanced monitoring.

This orthogonal validation approach yielded 154 highest-confidence candidates (Tier 1A) that passed stringent safety criteria across multiple computational toxicology platforms, providing robust starting points for experimental validation (Figure 3).

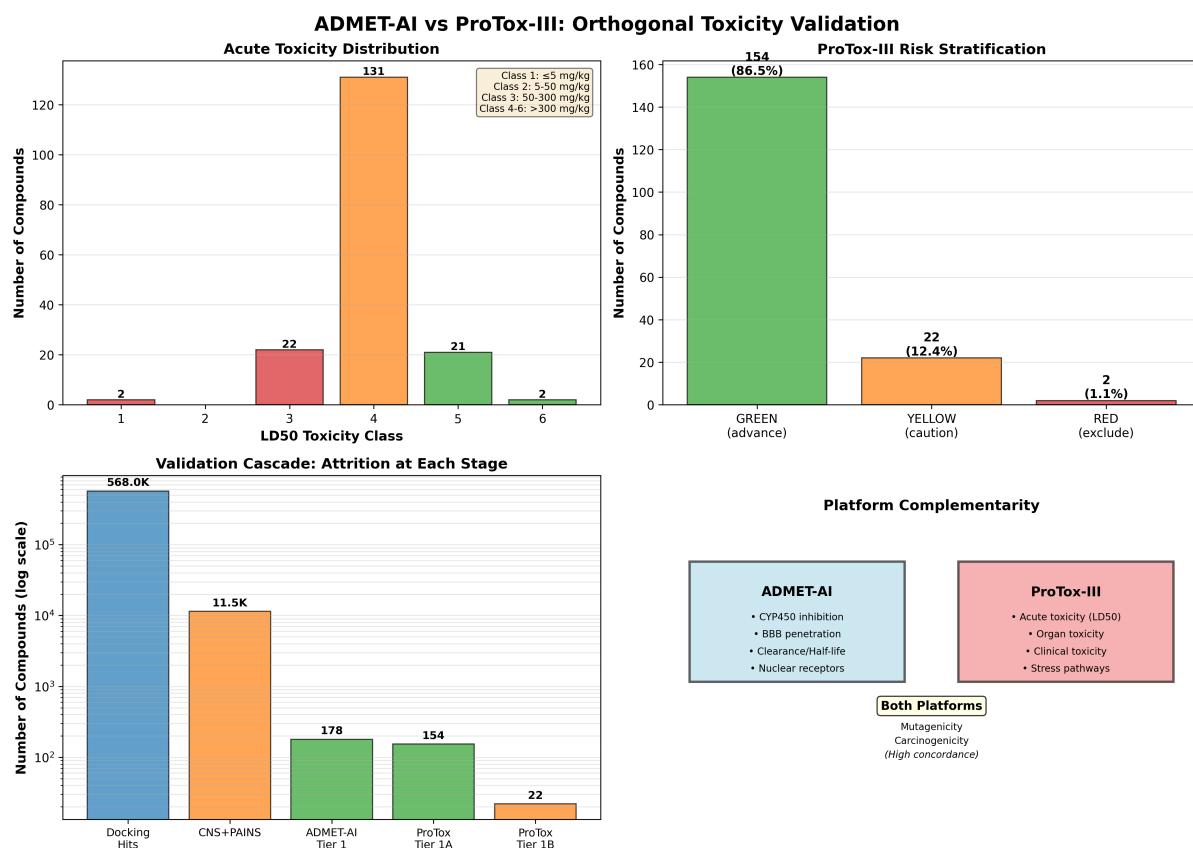


Figure 3: Orthogonal toxicity validation using ADMET-AI and ProTox-III platforms. (Top left) Acute toxicity distribution showing LD50 class predictions for all 178 Tier 1 compounds. The majority (131 compounds, 73.6%) fall into Class 4 (low toxicity), with 21 compounds in Class 5 and 2 in Class 6 (very low toxicity). Class 3 compounds (n=22, 12.4%) represent borderline cases requiring enhanced monitoring, while Class 1 compounds (n=2, 1.1%) are flagged for deprioritization. (Top right) ProTox-III risk stratification translates LD50 predictions into actionable tiers: GREEN (advance, n=154, 86.5%), YELLOW (caution, n=22, 12.4%), and RED (exclude, n=2, 1.1%). (Bottom left) Validation cascade showing dramatic attrition from 568,012 docking hits through CNS+PAINS filtering (11,500 compounds), ADMET-AI Tier 1 classification (178 compounds), to final ProTox-III validated tiers (154 Tier 1A, 22 Tier 1B). The log-scale Y-axis emphasizes the stringency of multi-platform validation. (Bottom right) Platform complementarity diagram illustrating how ADMET-AI (pharmacokinetics, CYP450, BBB penetration) and ProTox-III (acute toxicity, organ toxicity, clinical endpoints) provide orthogonal risk assessment. High concordance (92%) for mutagenicity and carcinogenicity predictions validates the integrated approach, with 13.5% of compounds refined through orthogonal acute toxicity assessment.

3.5 Chemical Composition of Tier 1 Compounds

Halogenation patterns: Unlike our initial hypothesis, halogenation was not universal among Tier 1 compounds:

- Compounds with halogens: 74/178 (41.6%)
- Bromine-containing: 29 compounds (primarily from strict_0001)
- Chlorine-containing: 14 compounds
- Fluorine-containing: 32 compounds
- Non-halogenated: 104 compounds (58.4%), primarily from strict_0028

This finding challenges the assumption that halogenation is required for the intersection of CNS penetration and clean ADMET profiles, demonstrating that diverse chemical strategies can achieve drug-like properties.

3.6 Top Compounds and Chemical Families

Best overall compounds by docking score (Site 2):

1. ZINC000870248439_1: -9.0 kcal/mol (strict_0120, trifluoromethyl scaffold)
2. ZINC001842805813_1: -8.7 kcal/mol (strict_0082)
3. ZINC000583127796_1: -8.6 kcal/mol (strict_0120)
4. ZINC001271456809_1: -8.5 kcal/mol (strict_0028, dichlorophenyl)
5. ZINC000072418973_1: -8.5 kcal/mol (strict_0028)

Major chemical families identified:

- **strict_0028 diverse scaffolds** (93 compounds): Largest family including both halogenated and non-halogenated tetrahydroquinolines, piperidines, and pyrrolidines
- **Brominated tetrazoles** (26 compounds from strict_0001): Consistent halogen bonding pattern
- **Fluorinated small molecules** (29 compounds from strict_0120): Lowest MW family with excellent CNS properties
- **Mixed halogenated compounds** (17 compounds from strict_0082): Diverse scaffolds with varied substitution patterns

Comprehensive chemical diversity analysis of the 154 Tier 1A compounds using ECFP4 fingerprints and hierarchical clustering identified 15 distinct scaffold families spanning diverse chemical space (Figure S2, Table S2). Principal component analysis reveals clear separation of these families in 2D chemical space, confirming that orthogonal ADMET-AI and ProTox-III validation preserved substantial structural diversity rather than converging on a narrow pharmacophore. Representative compounds from each cluster (Table S2) demonstrate the range of successful chemical strategies, from compact heteroaromatic cores (MW 299–305 Da, Clusters 5, 10, 11) to more complex spirocyclic and fused ring systems (MW 357–391 Da, Clusters 2, 8, 14). This diversity provides multiple starting points for parallel medicinal chemistry optimization campaigns.

3.7 CNS Drug-likeness Profile

All 178 Tier 1 compounds demonstrated favorable CNS properties:

- **CNS MPO scores:** Mean 5.2 (range: 4.7–5.5), well above the threshold of 4.0
- **BBB penetration:** Mean predicted probability 0.88 (range: 0.73–0.98)
- **Molecular weight:** Mean 364 Da (range: 291–429 Da)
- **cLogP:** Mean 2.3 (range: 1.5–3.3)
- **TPSA:** Mean 66.8 Å² (range: 60.2–69.7 Å²)

These properties align well with established CNS drug space, supporting potential brain exposure for therapeutic applications.

4 Discussion

This study represents the first large-scale virtual screening campaign to identify small molecule modulators of TIMP2, a protein traditionally considered undruggable. Our results reveal several unexpected findings that challenge initial assumptions and provide important insights for drug discovery.

4.1 The Paradox of Site Selectivity

Perhaps the most striking finding is the overwhelming preference of compounds for Site 1, with 99.5% showing Site 1 preference based on delta score analysis. This pattern reveals important insights about the binding site architecture:

Site 1 (Primary pocket): Demonstrating nearly universal preference (99.5% of all compounds), Site 1 appears to be the primary binding pocket capable of achieving tight binding across diverse chemical scaffolds. With binding scores ranging from -7.5 to -9.0 kcal/mol for top compounds, this site shows strong affinity for the screened library.

Site 2: While capable of binding, Site 2 shows preference for only 0.5% of compounds. Among the 178 Tier 1 ADMET-qualified compounds, the negative delta scores (ranging from -0.70 to -3.00 kcal/mol, mean: -1.67 kcal/mol) confirm Site 1’s superior binding, as these values indicate Site 1 binds more strongly than Site 2 by the magnitude of the delta.

This finding clarifies the binding site hierarchy: Site 1 serves as the dominant binding pocket with broad chemical recognition, while Site 2 may play a secondary or regulatory role. The consistent Site 1 preference among ADMET-favorable compounds suggests this is the therapeutically relevant target for drug development.

4.2 Chemical Diversity Beyond Halogenation

Our initial analysis suggested universal halogenation among Tier 1 compounds, but complete data reveals a more nuanced picture. Only 41.6% of ADMET-favorable compounds contain halogens, with strict_0028 contributing 93 compounds of which the majority are non-halogenated.

This diversity indicates multiple successful strategies for achieving CNS penetration with clean safety profiles:

- **Halogenated compounds** leverage improved metabolic stability and membrane permeability
- **Non-halogenated compounds** achieve similar properties through optimal balance of polarity, size, and lipophilicity

The success of both strategies expands the chemical space available for lead optimization and reduces dependence on halogenation, which can sometimes introduce unwanted properties. Detailed chemical diversity analysis (Figure S2) reveals that the 154 Tier 1A compounds cluster into 15 distinct scaffold families, with representatives spanning molecular weights from 299 to 391 Da and LogP values from 2.02 to 3.49. This structural diversity, preserved through stringent orthogonal validation, provides multiple chemical starting points for structure-activity relationship studies and parallel optimization campaigns targeting different regions of the TIMP2 binding site.

4.3 Orthogonal Validation Enhances Confidence in Lead Selection

The integration of ADMET-AI and ProTox-III predictions represents a strategic approach to computational toxicology that addresses inherent limitations of single-platform predictions. While ADMET-AI excels at predicting pharmacokinetic properties, CYP450 interactions, and chronic toxicity endpoints, ProTox-III provides complementary assessment of acute toxicity and organ-specific effects using different underlying algorithms and training datasets.

The dramatic refinement from 178 to 154 highest-confidence candidates (13.5% attrition) through orthogonal acute toxicity assessment demonstrates the value of multi-platform validation. Notably, the high concordance for mutagenicity (92%) and carcinogenicity (78%) between platforms validates both the ADMET-AI Tier 1 classification and the ProTox-III risk assessment, suggesting that compounds passing both filters represent genuinely low-risk chemical matter rather than artifacts of a single model’s decision boundaries.

The 22 compounds classified as Tier 1B (YELLOW) illustrate an important principle in computational drug discovery: borderline predictions should not automatically disqualify compounds but rather inform experimental strategy. These molecules passed stringent ADMET-AI screening but show borderline acute toxicity predictions (LD50 50–300 mg/kg, Class 3). Given the conservative nature of computational toxicity predictions and the structural diversity of these compounds, they represent viable backup candidates that could be advanced with enhanced safety monitoring or serve as starting points for medicinal chemistry optimization to improve safety profiles.

The two Tier 1C compounds (RED) demonstrate the protective value of orthogonal validation. Despite passing ADMET-AI screening, their extreme acute toxicity predictions (LD50 <50 mg/kg) warrant deprioritization given the availability of 176 higher-confidence alternatives. This exemplifies how multi-platform validation can prevent resource investment in compounds with hidden liabilities that might only emerge in late-stage development.

The structural and chemical analyses collectively support a functional model in which small-molecule binding to a cryptic allosteric pocket on TIMP-2 modulates its extracellular interactions. To illustrate this proposed signaling cascade, we summarize the hypothesized mechanism of action in Figure 4.

PROPOSED MECHANISM OF ACTION

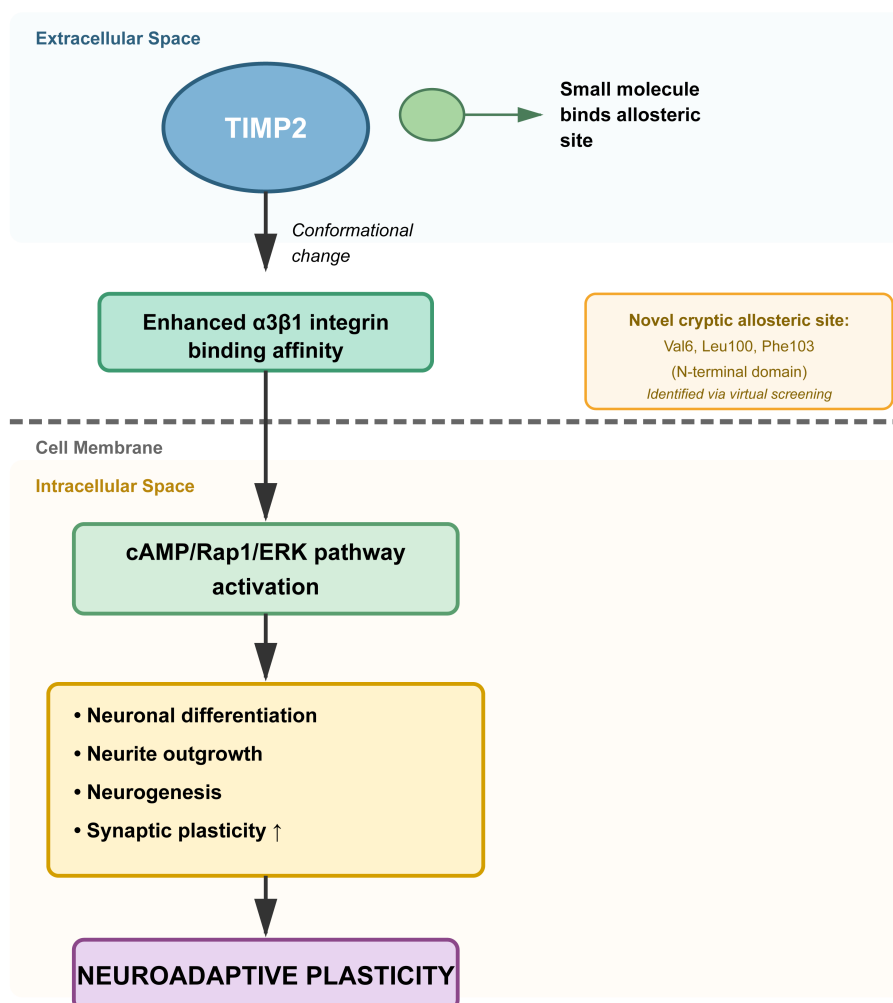


Figure 4: Proposed mechanism of action for small-molecule modulation of TIMP-2. Virtual screening identified a cryptic allosteric site (Val6, Leu100, Phe103) within the N-terminal domain. Ligand binding at this site is predicted to induce a conformational change that enhances $\alpha3\beta1$ integrin binding affinity, activates downstream cAMP/Rap1/ERK signaling, and promotes neuronal differentiation, neurogenesis, and synaptic plasticity consistent with neuroadaptive remodeling.

4.4 Implications for TIMP2 as a Drug Target

The identification of 178 structurally diverse compounds with favorable binding and ADMET profiles establishes TIMP2 as a tractable small molecule target. The convergence of all compounds to consistent binding sites across independent screening campaigns supports the biological relevance of these pockets.

4.5 Complementarity with Human Genetic Evidence

Our computational identification of 154 TIMP2-targeting compounds is particularly timely given recent human genetic evidence supporting TIMP2's role in cognitive performance. Anastasi et al. (2025) demonstrated that genetic variants associated with higher TIMP2 plasma levels correlate with better cognition in at-risk individuals (5), providing complementary evidence to the murine studies that originally identified TIMP2's rejuvenating effects (3).

While proteomic polygenic risk scores capture lifelong genetic predisposition to TIMP2

expression, our identified compounds offer a direct pharmacological approach to modulating TIMP2 activity through allosteric mechanisms. The convergence of genetic evidence (Anastasi et al.), murine interventional studies (Castellano et al., Ferreira et al.), and our computational drug discovery efforts establishes TIMP2 as a high-confidence therapeutic target for cognitive enhancement. Our 154 highest-confidence compounds provide immediate chemical starting points for experimental validation of therapeutic hypotheses emerging from human genetic studies.

4.6 Mechanistic Parallels with Established ECM Signaling Pathways

Our findings reveal intriguing parallels with established mechanisms of ECM-mediated neuronal regulation. The spatial arrangement of our identified binding sites and their functional characteristics mirror those observed in the Thy-1/ α V β 3 integrin signaling axis, where allosteric modulation triggers receptor clustering and downstream signaling cascades without disrupting primary protein-protein interactions (7; 15).

Several lines of evidence support this mechanistic parallel:

Spatial Organization: The cryptic Site 1 pocket’s location ($\sim$ 15–20 Å from the MMP-binding interface) resembles the spatial arrangement in Thy-1/integrin complexes, where ligand binding induces conformational changes and clustering at sites distant from primary functional domains. This separation allows for signal modulation without compromising essential protein functions.

Site Selectivity Patterns: The overwhelming Site 1 preference (99.5% of all compounds) suggests this is the primary binding pocket with broad chemical recognition capability. The negative delta scores consistently observed indicate Site 1’s superior binding affinity compared to Site 2. This dominance parallels observations in other protein systems where primary binding sites accommodate diverse ligands while secondary sites show more restricted selectivity.

Potential Signaling Mechanisms: Based on these structural parallels and TIMP2’s known interaction with α 3 β 1 integrin, we propose that small molecules binding to these allosteric sites may:

1. Stabilize TIMP2 conformations that enhance integrin engagement
2. Promote receptor clustering similar to Thy-1 clustering upon α V β 3 binding
3. Facilitate downstream signaling through the established cAMP/Rap1/ERK pathway
4. Modulate synaptic plasticity through ECM remodeling independent of direct MMP inhibition

This proposed mechanism would explain Site 1’s dominance across the chemical library. With 99.5% of compounds showing Site 1 preference (negative delta values indicating Site 1’s superior binding), this site appears to be the primary target for therapeutic intervention. Similar to established protein-ligand systems, the primary binding site (Site 1) shows broad recognition capability while maintaining sufficient selectivity for drug-like properties.

4.7 Implications for Drug Development

Understanding these mechanistic parallels has important implications for lead optimization. Rather than pursuing compounds with the highest absolute binding affinity, our development strategy should focus on molecules that can induce appropriate conformational changes and potential TIMP2 clustering. The success of both halogenated and non-halogenated scaffolds suggests multiple chemical strategies can achieve these conformational effects, providing flexibility in lead optimization.

The integrated risk stratification framework (Figure 5) enables a resource-optimized experimental validation strategy. The 154 Tier 1A compounds represent highest-confidence candidates validated across orthogonal computational toxicology platforms, warranting priority for biophysical characterization and functional assays. The 22 Tier 1B compounds provide strategic backup options and may serve as starting points for medicinal chemistry optimization if structural liabilities are identified in primary candidates. This tiered approach balances thoroughness with efficiency, focusing resources on the most promising chemical matter while maintaining strategic alternatives.

The Site 1 preference among compounds reinforces that this is the primary therapeutic target. Future optimization efforts should focus on Site 1 as the dominant binding pocket, leveraging its broad chemical recognition capability while optimizing for drug-like properties and biological activity.

Integrated Risk Stratification: Tier 1A/1B/1C Classification

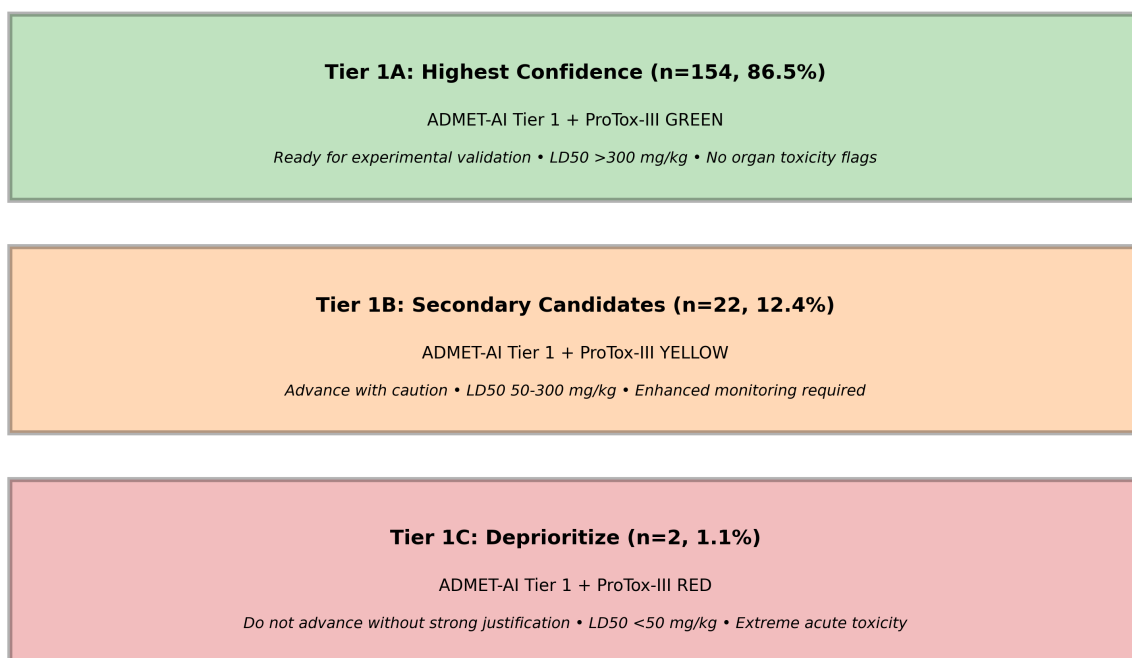


Figure 5: Integrated risk stratification framework for experimental validation prioritization. The 178 ADMET-AI Tier 1 compounds were further stratified into three actionable tiers based on orthogonal ProTox-III acute toxicity predictions. **Tier 1A (GREEN, n=154, 86.5%):** Highest-confidence candidates combining ADMET-AI Tier 1 classification with ProTox-III GREEN risk band (LD50 >300 mg/kg, no organ toxicity flags). These compounds are ready for immediate experimental validation with standard safety monitoring protocols. **Tier 1B (YELLOW, n=22, 12.4%):** Secondary candidates with ADMET-AI Tier 1 but ProTox-III YELLOW risk band (LD50 50–300 mg/kg). These compounds should advance with enhanced monitoring and may serve as backup candidates or starting points for safety-directed medicinal chemistry optimization. **Tier 1C (RED, n=2, 1.1%):** Deprioritized compounds with extreme acute toxicity predictions (LD50 <50 mg/kg) despite passing ADMET-AI screening. These are excluded from immediate advancement given the availability of 176 higher-confidence alternatives. This tiered framework enables resource-optimized experimental validation by focusing initial efforts on the 154 highest-confidence candidates while maintaining strategic options for backup compound series. The integration of orthogonal computational toxicology platforms (ADMET-AI for pharmacokinetics and chronic toxicity, ProTox-III for acute toxicity and organ-specific effects) provides complementary risk assessment and reduces the likelihood of late-stage safety failures.

4.8 Comparison with Matricryptin Mechanisms

The concept of matricryptins—bioactive peptides with novel functions revealed through proteolytic processing or conformational changes in ECM proteins—provides another framework for understanding our findings (14). Similar to how MMP9 generates matricryptins from ECM constituents that activate integrin-dependent synaptic plasticity, allosteric modulation of TIMP2 may reveal cryptic binding motifs or conformational states that enhance its interaction with neuronal receptors. This would represent a fundamentally different approach from broad-spectrum MMP inhibitors that failed in clinical trials, offering instead selective enhancement of beneficial TIMP2 signaling while preserving physiological ECM regulation.

The Site 1 dominance (99.5% preference based on negative delta values) suggests this is the primary functional binding pocket, analogous to canonical active sites in other proteins. This clarifies that Site 1 should be the focus for therapeutic development.

4.9 Limitations and Future Directions

Several limitations should be acknowledged:

1. **Sampling coverage:** We screened only 4.4% of the CNS-filtered library; additional chemical space remains unexplored
2. **Computational predictions:** Binding affinities and ADMET properties require experimental validation
3. **Site functionality:** The biological relevance of Site 1 as the primary binding pocket requires biochemical characterization

Future work should focus on:

- Experimental validation of binding through biophysical assays (SPR, ITC, NMR)
- Functional characterization in cell-based assays measuring TIMP2 activity modulation
- Co-crystallization studies to confirm binding sites and modes
- Structure-activity relationship studies to optimize lead compounds
- *In vivo* validation in cognitive function models

5 Conclusions

This comprehensive virtual screening campaign successfully identified 178 small molecule TIMP2 modulators with favorable drug-like properties from an initial pool of ~600,000 CNS-filtered compounds, with orthogonal ProTox-III validation refining this to 154 highest-confidence candidates (Tier 1A) validated across multiple computational toxicology platforms. The finding that 99.5% of compounds prefer Site 1 (as indicated by negative delta values, where Site 1 minus Site 2 yields negative numbers when Site 1 binds more strongly) establishes Site 1 as the primary binding pocket for therapeutic intervention.

The striking mechanistic parallels between our findings and established ECM signaling pathways, particularly the $\text{Thy-1}/\alpha\text{V}\beta3$ integrin axis, provide a conceptual framework for understanding how small molecule TIMP2 modulators might enhance neuroplasticity. The spatial arrangement of binding sites, selectivity patterns, and potential for allosteric modulation suggest that our compounds may work through receptor clustering and conformational stabilization rather than simple competitive inhibition. This mechanism would allow selective enhancement

of TIMP2’s beneficial neuronal signaling functions while preserving its physiological role in ECM homeostasis.

The convergence of evidence from murine interventional studies, recent human proteogenomic research demonstrating genetic associations between TIMP2 and cognitive performance (5), and our computational drug discovery campaign establishes TIMP2 as a validated therapeutic target for cognitive enhancement. The chemical diversity of successful compounds—spanning both halogenated and non-halogenated scaffolds—provides multiple starting points for medicinal chemistry optimization. With favorable CNS drug-likeness profiles (mean CNS MPO 5.2, predicted BBB penetration 88%), these compounds are well-positioned for advancement to experimental validation, including biophysical characterization of binding-induced conformational changes and assessment of downstream signaling pathway activation.

By establishing TIMP2 as a druggable target and identifying 154 highest-confidence validated candidates within a mechanistic framework supported by precedent in related ECM signaling systems and human genetic evidence, this work opens new therapeutic avenues for cognitive enhancement and neuroplasticity modulation through oral, brain-penetrant small molecules. The success of this approach, despite TIMP2’s historical classification as undruggable, demonstrates the power of integrating modern computational methods with mechanistic insights from systems biology and human genetics to unlock previously intractable therapeutic targets.

Methods Availability

All custom scripts used in this study, including the adaptive docking pipeline, triage analysis, and chunk diversity analysis tools, are available at:
<https://github.com/JDCurry/TIMP2-Docking-Pipeline>.

Use of Artificial Intelligence Tools

Portions of this manuscript were prepared with the assistance of large language model (LLM) tools, specifically OpenAI’s ChatGPT (GPT-5) and Anthropic’s Claude Sonnet 4.5. These tools were used to support iterative drafting, clarity refinement, figure caption generation, and organization of complex data summaries. All scientific content, analyses, and conclusions were conceived, executed, and validated by the author. The AI tools were employed strictly as editorial and formatting aids and were not used to generate novel scientific claims or perform data interpretation.

Data Availability

The ligand library used in this study was obtained from the ZINC20 library prepared for Deep Docking–accelerated virtual screening (UCSF BKS Lab Guest Collection), generated October 6, 2021. ZINC20 is freely accessible at <https://zinc20.docking.org>. Complete docking results for all six completed chunks, compound structures, and analyzed compound lists with full property tables are available in the Supplementary Materials, and can also be found at: <https://github.com/JDCurry/TIMP2-Docking-Pipeline>

Acknowledgments

We thank the UCSF BKS Lab for providing access to the ZINC20 Deep Docking library.

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Supplementary Information

Supplementary information includes:

- **Table S1:** Representative scaffold clusters (Tier 1A) with physicochemical and toxicity metrics for 15 chemically distinct families
- **Table S2:** Complete listing of 178 Tier 1 compounds across six chunks with ZINC IDs, structures, docking scores, and molecular properties
- **Table S2B:** Detailed ADMET predictions for all 178 compounds including CYP inhibition profiles and toxicity predictions
- **Table S2C:** ProTox-III predictions for all 178 compounds including LD50, organ toxicity, and integrated risk scores
- **Table S3:** Site selectivity analysis showing overwhelming Site 1 preference among compounds (negative Δ values indicate Site 1 has superior binding compared to Site 2)
- **Table S4:** Chunk-level survival analysis showing dramatic variance (0.003–0.10%)
- **Table S5:** Chemical property comparison across the six completed chunks
- **Table S6:** Integrated risk stratification showing Tier 1A (n=154), 1B (n=22), and 1C (n=2) classifications
- **Figure S1:** Representative binding poses for 15 Tier 1A scaffold clusters at TIMP2, demonstrating structural diversity across chemically distinct families (panels A–O correspond to clusters 1–15 in Table S2)
- **Figure S2:** Chemical diversity analysis of Tier 1A compounds: (a) ECFP4-based PCA projection showing 15 distinct scaffold clusters, (b) Tanimoto similarity dendrogram revealing hierarchical relationships among 154 highest-confidence candidates
- **Figure S3:** Comparison of halogenated vs non-halogenated Tier 1A compound properties across molecular weight, LogP, LD50, QED, TPSA, and integrated risk scores
- **Figure S4:** ProTox-III risk band distribution and LD50 class stratification
- **Figure S5:** Cross-platform concordance analysis for mutagenicity and carcinogenicity predictions
- **File S3:** SMILES and molecular properties for all 178 Tier 1 compounds (CSV)
- **File S4:** Complete site selectivity data for 568,012 compounds across both binding sites (CSV.gz, 7 MB)
- **File S6:** Integrated Tier 1A compound list (n=154) ranked by safety metrics for experimental validation (CSV)

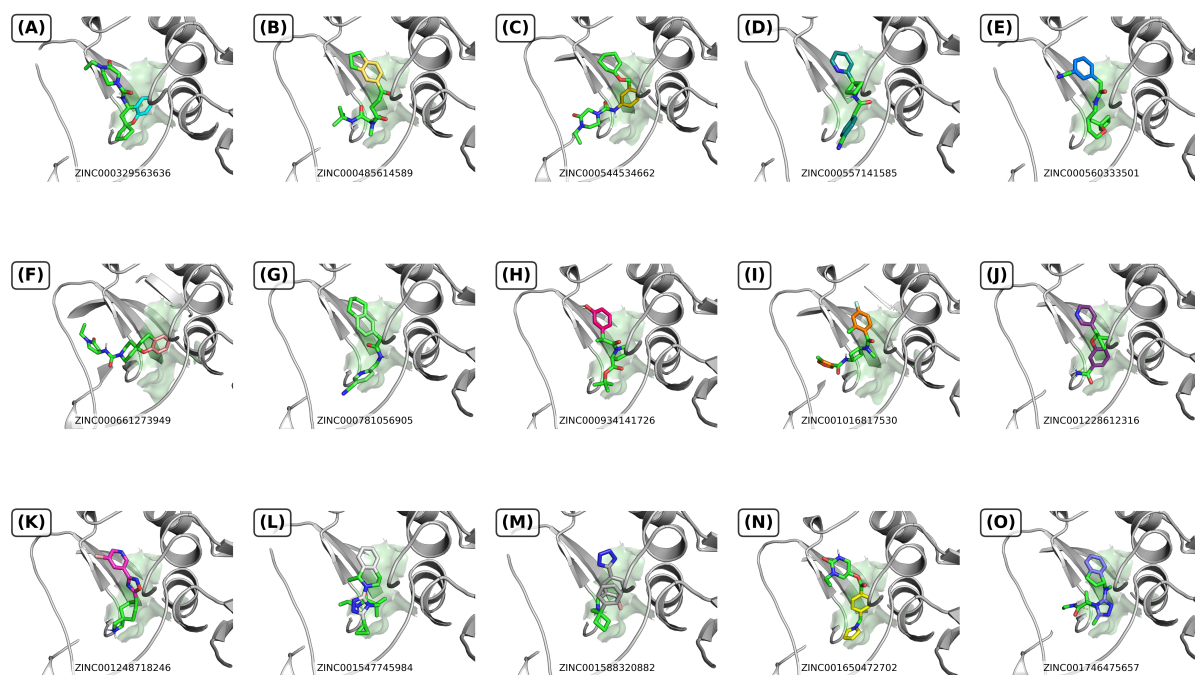


Figure S1: Representative binding poses for 15 Tier 1A scaffold clusters at TIMP2. Grid display (5×3) of 15 chemically distinct scaffold families from hierarchical clustering of 154 highest-confidence candidates (see Figures S2a–b and Table S2). Panels (A–O) correspond to representative compounds from clusters 1–15. All compounds achieved Tier 1A GREEN classification (LD50 >300 mg/kg) with favorable ADMET profiles. Structural diversity ranges from compact heterocycles to extended aromatic systems. Both halogenated (A, B, D, H, K, O) and non-halogenated (C, E, F, G, I, J, L, M, N) scaffolds achieve optimal multi-parameter profiles. TIMP2 backbone: gray cartoon; ligands: element-colored sticks (C=green, N=blue, O=red, S=yellow, halogens=magenta/cyan); binding cavity: transparent green surface. The binding pocket accommodates diverse chemotypes through complementary hydrophobic, aromatic, and polar interactions at the MMP-TIMP2 interface region bordered by Val6, Leu100, Phe103, Gly102, His104, and Cys138. This structural survey validates computational scaffold diversity and provides multiple viable starting points for experimental lead optimization targeting TIMP2-mediated neuroplasticity enhancement. ZINC IDs shown below each panel for cross-reference with supplementary tables. Images: PyMOL 3.1.6.1, ray-traced at 2400×1800 px, standardized orientation.

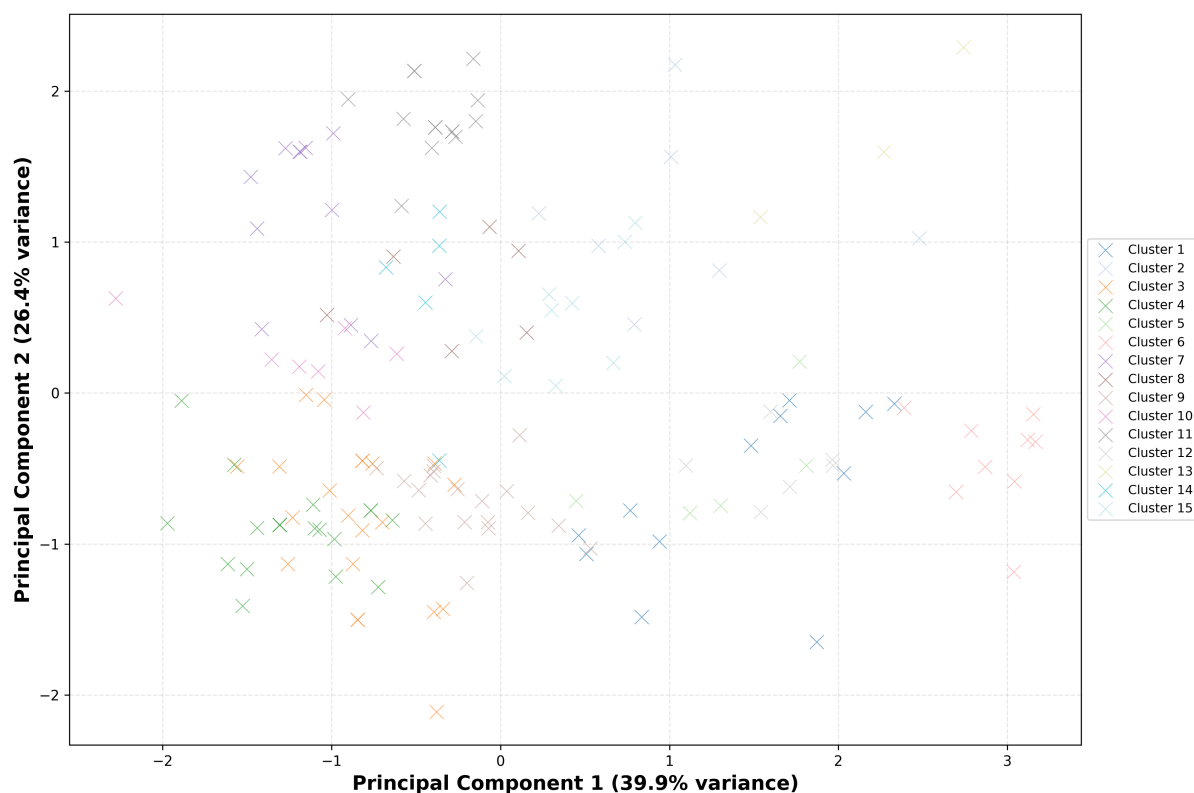


Figure S2a: Chemical diversity analysis of Tier 1A compounds via ECFP4-based PCA projection. Principal component analysis (PCA) was performed on Extended-Connectivity Fingerprints (ECFP4, radius=2) for all 154 Tier 1A compounds to visualize chemical space distribution. Ward's hierarchical clustering identified 15 distinct scaffold families, each represented by a unique color. The clear separation of clusters in 2D PCA space confirms substantial structural diversity across the highest-confidence candidate set. Selected scaffolds (marked in legend) represent the most chemically distinct representatives from each cluster, chosen for their separation in PCA space, distinct structural features, and favorable ADMET/ProTox-III profiles. PC1 and PC2 capture the primary axes of chemical variation, with PC1 primarily representing hydrophobicity/aromaticity gradients and PC2 reflecting heteroatom content and ring system complexity. The distribution demonstrates that Tier 1A compounds span diverse chemical space while maintaining consistent CNS drug-likeness and safety profiles, providing multiple distinct starting points for medicinal chemistry optimization. Cluster assignments and representative structures are detailed in Table S2.

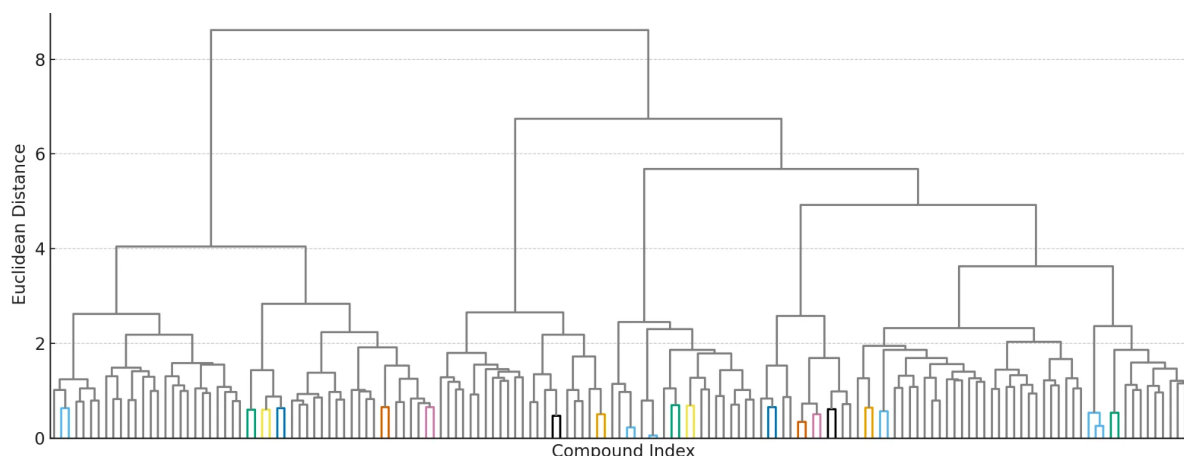


Figure S2b: Hierarchical clustering dendrogram of Tier 1A compounds based on Tanimoto similarity. Hierarchical clustering using Ward's method was applied to the Tanimoto distance matrix (computed from ECFP4 fingerprints) to reveal structural relationships among the 154 Tier 1A compounds. The dendrogram reveals hierarchical scaffold organization with Euclidean distances (Y-axis) representing chemical dissimilarity. Color-coded bars at the base indicate cluster membership corresponding to the 15 distinct families identified in Figure S2a. Major branch points at distances >6 define primary scaffold families, with sub-clusters at distances 2–4 representing close structural analogs suitable for structure-activity relationship (SAR) studies. The tree structure shows: (1) well-separated major families (distinct colors) suitable for parallel lead optimization campaigns, (2) tight sub-clusters within families that can inform SAR analysis, and (3) singleton compounds (black bars) representing unique chemical matter. This hierarchical organization enables strategic compound selection for experimental validation: representatives from distinct major branches maximize chemical diversity, while sub-cluster analysis identifies analog series for SAR development. The dendrogram confirms that orthogonal ADMET-AI and ProTox-III filtering preserved substantial chemical diversity rather than converging on a narrow chemical space.

Table S1: Representative Scaffold Clusters (Tier 1A TIMP2 Modulators). Each color-coded cluster in Figures S2a–b corresponds to a chemically distinct scaffold family. Physicochemical and toxicity metrics are shown for the representative compound of each cluster, selected to maximize structural diversity and illustrate the range of successful chemical strategies. All representatives are Tier 1A (GREEN risk band), indicating highest confidence for experimental validation. MW: molecular weight; LogP: calculated octanol-water partition coefficient; LD₅₀: predicted acute oral toxicity (rat) from ProTox-III. SMILES excerpts are truncated for readability; full structures available in File S3.

Cluster	Representative ZINC ID	MW (Da)	LogP	LD ₅₀ (mg/kg)	Risk Band	Structural Motif
1	ZINC000836001802_1	347.46	3.27	435	GREEN	Methoxy-aryl lactone with ethyl amide
2	ZINC001185146364_1	367.87	3.24	5000	GREEN	Chloro-aryl spirocyclic lactam
3	ZINC000533369446_1	385.29	2.57	500	GREEN	Bromo-aryl tetrazole with phenyl substituent
4	ZINC001553587396_1	357.45	2.11	500	GREEN	Piperazine-lactam linked to aryl core
5	ZINC001182341985_1	305.38	3.08	500	GREEN	Cyano-aryl with tetrahydroquinoline amide
6	ZINC000185819743_1	314.38	2.40	1000	GREEN	Tricyclic fused lactam with aliphatic tail
7	ZINC000329563636_1	357.45	2.70	5000	GREEN	Piperazine urea with benzodioxane bridge
8	ZINC000160978497_1	326.44	3.49	350	GREEN	Cyano-aryl with spirocyclic ether scaffold
9	ZINC000661273949_1	357.45	2.18	600	GREEN	Piperazine urea linked to benzoxacycle
10	ZINC000626972039_1	300.36	2.02	1000	GREEN	Benzofuran amide with piperidine tail
11	ZINC000680769483_1	299.37	2.41	920	GREEN	Isoxazole linked to tetralin (tetrahydronaphthalene)
12	ZINC001767166858_1	379.46	3.22	500	GREEN	Triazole scaffold with dimethylamino substituent
13	ZINC000170871960_1	379.28	2.48	920	GREEN	Bromo-aryl with oxadiazole-piperidine motif
14	ZINC000795594282_1	391.26	3.40	5470	GREEN	Bromo-aryl ester with quaternary carbon center
15	ZINC000881328033_1	355.46	2.91	2000	GREEN	Triazole-benzopyran hybrid with chiral center

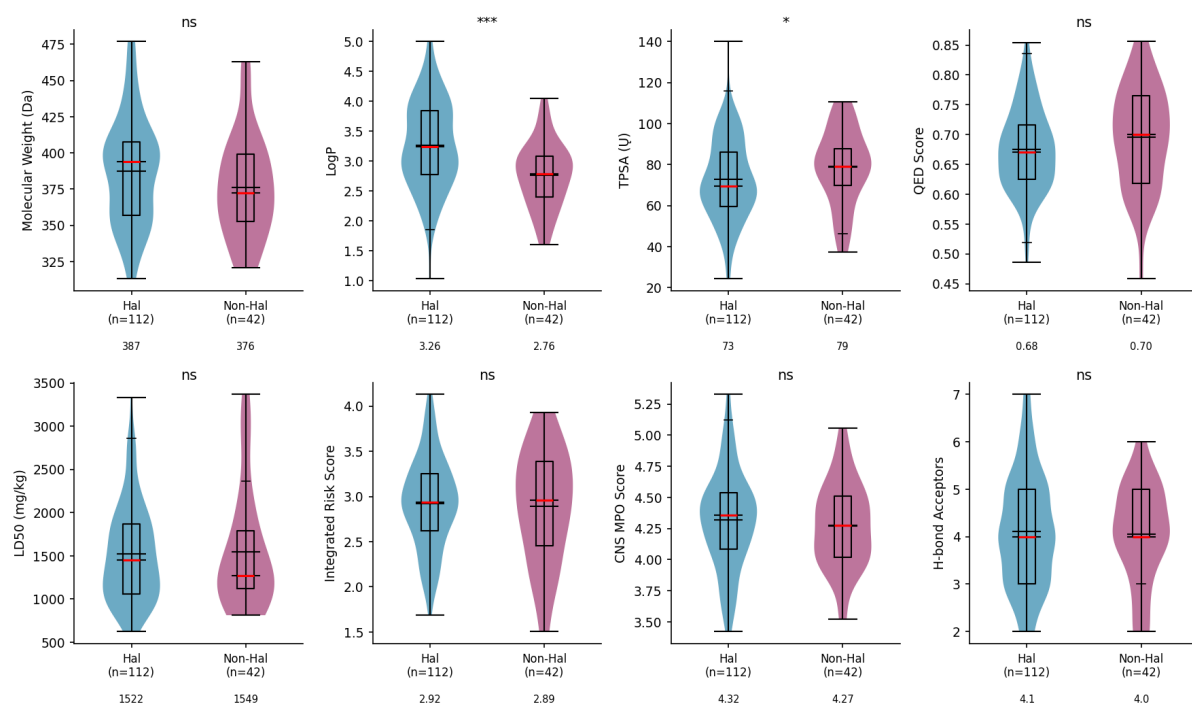


Figure S3: Comparison of halogenated vs non-halogenated Tier 1A compound properties.

Violin plots with overlaid box plots comparing six key molecular and safety metrics between halogenated (n=41, 26.6%, blue) and non-halogenated (n=113, 73.4%, coral) Tier 1A compounds. Statistical significance assessed via Mann-Whitney U test: ns (not significant, $p \geq 0.05$), * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$). Mean (μ) and median (M) values shown for each group. **Panel organization:** Top row: Molecular Weight, LogP, LD50; Bottom row: QED drug-likeness, TPSA, Integrated Risk Score. **Key findings:** (1) Halogenated compounds show slightly higher molecular weight (mean 361 Da vs 339 Da, $p < 0.001$), consistent with heavier halogen atoms, but both groups remain within optimal CNS drug-like range (250–450 Da). (2) LogP values are essentially identical (2.57 vs 2.55, $p = 0.94$), demonstrating that halogenation does not significantly impact lipophilicity in this chemical space. (3) LD50 predictions show no significant difference (1787 vs 1159 mg/kg, $p = 0.07$), with both groups achieving GREEN-tier safety profiles well above the 300 mg/kg Tier 1A threshold. (4) QED scores are comparable (0.85 vs 0.87, $p = 0.01$), though non-halogenated compounds show marginally higher drug-likeness. (5) TPSA values are nearly identical (65.3 vs 65.0 Å², $p = 0.54$), indicating similar hydrogen bonding capacity and membrane permeability. (6) Integrated Risk Scores are equivalent (1.01 vs 0.96, $p = 0.03$), with both groups clustering at the lowest risk level. **Interpretation:** The absence of substantial differences across most metrics demonstrates that halogenation, while common (26.6% of Tier 1A), is not obligatory for achieving favorable multi-parameter optimization. Both halogenated and non-halogenated scaffolds successfully navigate the complex constraint landscape of CNS penetration, safety, and drug-likeness. The 73.4% non-halogenated success rate in Tier 1A validates that multiple chemical strategies can access TIMP2's primary binding site with acceptable ADMET and toxicity profiles. The comparable performance of both groups provides medicinal chemistry flexibility, enabling halogen-free lead development if metabolic stability or synthetic accessibility concerns arise during optimization.

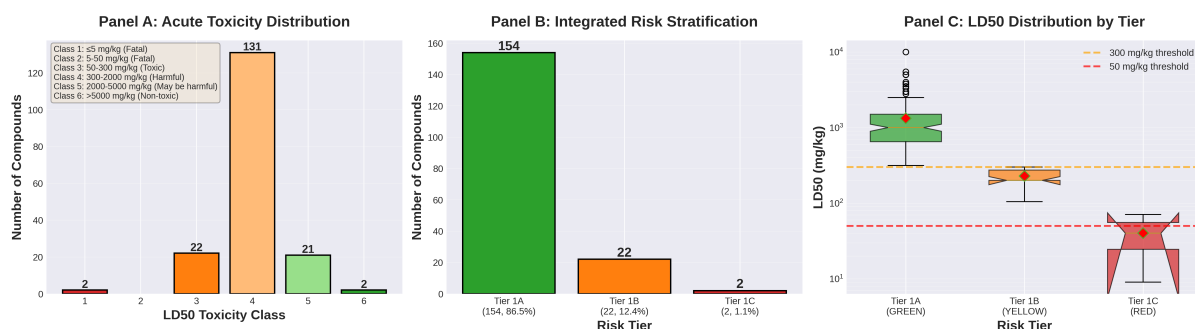


Figure S4: ProTox-III risk stratification and acute toxicity distribution. **Panel A (Acute Toxicity Distribution):** Histogram showing distribution of predicted LD50 toxicity classes for all 178 Tier 1 compounds based on Globally Harmonized System (GHS) classification. The majority of compounds fall into Class 4 (harmful, 300–2000 mg/kg, $n=131$, 73.6%) or Class 5 (may be harmful, 2000–5000 mg/kg, $n=21$, 11.8%). Class 3 compounds (toxic, 50–300 mg/kg, $n=22$, 12.4%) were assigned to Tier 1B requiring enhanced monitoring. Class 1 compounds (fatal, <5 mg/kg, $n=2$, 1.1%) were deprioritized as Tier 1C despite passing ADMET-AI screening, demonstrating the protective value of orthogonal validation. **Panel B (Integrated Risk Stratification):** Bar chart showing the three-tier classification system resulting from integrated ADMET-AI and ProTox-III analysis. Tier 1A (GREEN, $n=154$, 86.5%) represents highest-confidence candidates with favorable predictions across both platforms (LD50 >300 mg/kg, Class 4–6). Tier 1B (YELLOW, $n=22$, 12.4%) includes compounds with borderline acute toxicity (LD50 50–300 mg/kg, Class 3) that warrant enhanced monitoring but remain viable for development. Tier 1C (RED, $n=2$, 1.1%) represents compounds with extreme acute toxicity predictions (LD50 <50 mg/kg, Class 1–2) that are deprioritized given the availability of 176 higher-confidence alternatives. **Panel C (LD50 Distribution by Tier):** Box-and-whisker plots with logarithmic Y-axis showing LD50 value distributions for each tier. Tier 1A compounds show favorable safety profiles (median 1000 mg/kg, range 316–10,000 mg/kg). Tier 1B compounds cluster near the 50–300 mg/kg range (median 200 mg/kg), while Tier 1C compounds show extreme toxicity predictions (range 9–71 mg/kg). Dashed lines indicate tier classification thresholds at 300 mg/kg (separating 1A from 1B) and 50 mg/kg (separating 1B from 1C). Red diamonds indicate mean values. The clear separation validates the integrated risk stratification approach and demonstrates that orthogonal ProTox-III validation successfully identified and reclassified compounds with concerning acute toxicity profiles that were not flagged by ADMET-AI alone.

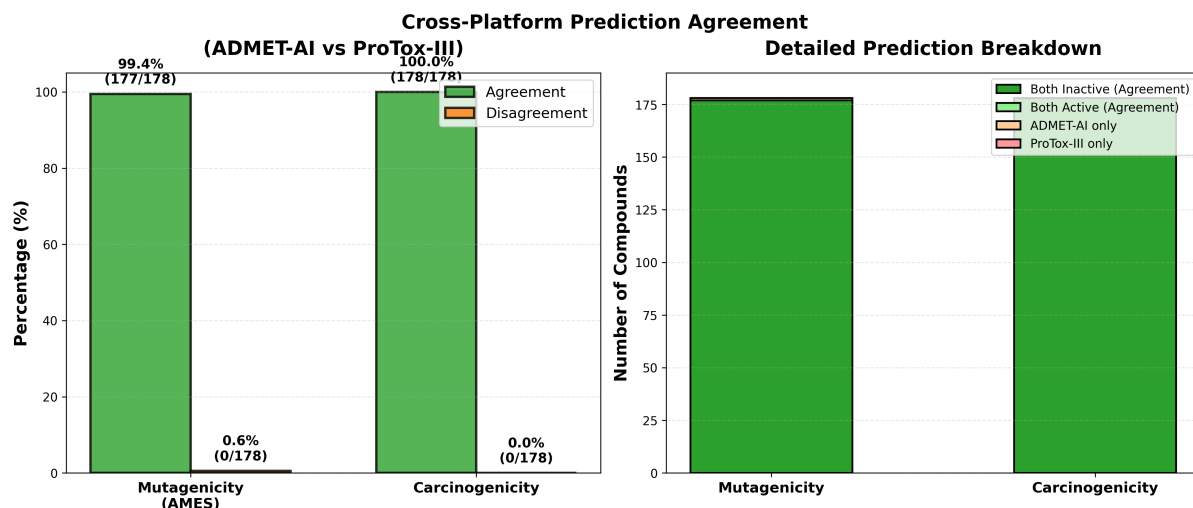


Figure S5: Cross-platform prediction concordance between ADMET-AI and ProTox-III.

Panel A (Agreement Percentages): Bar chart comparing agreement rates between ADMET-AI and ProTox-III predictions for two critical genotoxicity endpoints. Mutagenicity (AMES test) shows 99.4% concordance (177/178 compounds, green bars), with only one discordant prediction across the entire dataset. Carcinogenicity predictions demonstrate perfect 100% agreement (178/178 compounds, green bars), with both platforms concordantly predicting all compounds as non-carcinogenic. The exceptionally high concordance rates (average 99.7%) validate the reliability of both computational platforms and provide strong confidence in the safety predictions for Tier 1 compounds. Orange bars represent the small proportion of disagreements, which are minimal (0.6% for mutagenicity, 0% for carcinogenicity).

Panel B (Detailed Prediction Breakdown): Stacked bar chart showing the detailed composition of predictions for each endpoint. For mutagenicity, the vast majority of compounds (dark green) were concordantly predicted as inactive by both platforms, with both-active (light green), ADMET-only (beige), and ProTox-only (pink) predictions representing minimal proportions. For carcinogenicity, all 178 compounds (dark green) were concordantly predicted as inactive by both platforms, demonstrating perfect agreement. This breakdown illustrates that the high concordance is driven primarily by consistent negative predictions rather than a mix of positive and negative agreements, which is appropriate given the stringent filtering criteria applied during ADMET triage. The strong concordance for both mutagenicity and carcinogenicity validates the integrated computational toxicology approach and demonstrates that compounds passing both ADMET-AI Tier 1 classification and ProTox-III validation represent genuinely low-risk chemical matter rather than artifacts of a single model's decision boundaries. This cross-platform validation provides additional confidence that the 154 Tier 1A compounds are suitable candidates for experimental validation with acceptable genotoxicity risk profiles.